

GC University Lahore

Department of Computer Science

Examination: Fall 2015

Course: Database Systems

Course Code: CS-2014

Semester: V

Allowed Time: 155 Minutes

Total Marks: 56

PART A: Objective Type Questions

Total Marks: 14

Note: Encircle the right answer. Erasing/cutting or overwriting shall bear "No Marks".

1. Metadata is:

- ☐ a) Syntax of data
- ☐ b) Semantic of data
- ☐ c) Data about data
- ☐ d) Multiple data

2. A conceptual schema is:

- ☐ a) User view of the data
- ☐ b) DBMS view of the data
- ☐ c) Analyst view of the data
- ☐ d) None of the above

3. An entity relating to itself by an ER diagram is called:

- ☐ a) Recursive relationship
- ☐ b) Unary relationship
- ☐ c) Both (a) & (b)
- ☐ d) None of (a) & (b)

4. _____ refers to a database modification/distribution in which different portion of the database reside at different nodes of network.

- a) Replication
- b) Partitioning
- c) Application
- d) Binding

5. In three level schema architecture, Entity Relationship diagram is produced as:

- a) External level
- b) Internal level
- c) Conceptual level
- d) None of the above

6. The development of ER diagram is a:

- a) Linear Process
- b) Sequential Process
- c) Iterative process
- d) static process

7. An Entity Type is defined as:

- a) A set of relationships
- b) A set of attributes
- c) A combination of entities and relationship
- d) A set of entities sharing common characteristics

8. The anomalies addressed by moving from 3NF to BCNF deals with:

- a) Extensive updates and redundancy of data for each entity
- b) Inability to uniquely identify an entity
- c) Inability to reconstruct the relations once they are broken
- d) Creation of identical rows in a relation

9. The manipulated and processed data is called:

- a) Knowledge
- b) Information

- c) Metadata
- d) Graph

10. Let $R(A,B,C,D)$ be a relation in Boyce-Codd Normal Form (BCNF). Suppose ABC is the only key for R . Which of the following functional dependencies is guaranteed to hold for R ?

- a) $D \rightarrow A$
- b) $BC \rightarrow A$
- c) $ABCD \rightarrow E$
- d) None of these

11. If the relationship is of optional between two related entities, the resultant relational model would have:

- a) Foreign key as Not Null
- b) Foreign key as part of primary key
- c) Foreign key as Null
- d) None of the above

12. Which clause is Select statement execute before any other clause:

- a) Order by clause
- b) From clause
- c) Where clause
- d) Group by clause

13. Data redundancy means:

- a) Duplication
- b) Uniqueness
- c) Consistent
- d) Security

14. Which of the following is an example of one-to-one relation:

- a) Mother - Daughter
- b) Country - Capital
- c) Person - Automobile

- d) Doctor - Patient

PART B: Subjective Questions

Total Marks: 56

Q No. 1.

- **a) What is Normalization? Explain 2NF, 3NF and BCNF with examples. (7)**
- **b) Write down the advantages of Database Systems. (3)**

Q No. 2.

- **a) What are DDL, DRL and DML, give detail. (3)**
- **b) A company desires to keep record of multiple addresses for their employees, What kind of attribute is the solution to this problem. Also Model this scenario with respect to ERD and Normalization. (4)**
- **c) What are three main integrity constraints? Define Relationship and degree of relationship with an example. (7)**

Q No. 3.

- **a) Suppose a University which has a number of libraries, to support students & staff in providing study materials, each library has its identification number and name. These libraries are located in different buildings. Each library is managed by a librarian. The librarian also acts as an officer to other library staff. Both librarian and staff are considered employees of the University, employees are identified by employee_id, name, address and phone numbers. Books are written by authors, authors have id's and Name. An author can write many books and a book could be written by many authors. Each book belongs to a specific subject. An author can write on any number of subjects but at least one. The books are identified by ISBN number and they have book life and year of publication. Many copies of each book are placed in the library but at least one copy is always there. These copies are catalogued, and placed on different shelves. A librarian can issue a copy of book to member of the library, and define issue date and return date. If the book is not returned on specified date then the member is fined. (10)**
 1. Draw an Entity-Relationship diagram with primary and foreign keys, also allocated normalized tables in relations in your ERD.
 2. Additionally requirements may be generated for illustration purposes, however, write them down for further verification.
- **b) Explain Optional and Mandatory Cardinality in a relationship with example. (4)**

Q No. 4.

- **a) Convert following table into 2NF and 3NF: (10)**

Student_ID	Major	Student_name	Course_Code	Course_Title	Attendance Percentage
100-BSCS-13	CS	Hamza	CS-2014	Database Systems	90%
100-BSCS-13	CS	Hamza	CS-3206	Data Mining	85%
101-BSCS-13	IT	Kashif	CS-2014	Database Systems	82%

Note: Write down appropriate functional dependencies with each step of normalization. Explain the process by highlighting the important points as you proceed with the Normalization.

Q No. 5.

- **a) Define transaction? Also elaborate properties of transaction. (4)**
- **b) Elaborate lost updates, uncommitted data and inconsistent retrievals when simultaneous transactions are executed on a shared database. (3)**
- **c) Explain ETL and discuss the types of extraction, transformation and loading in detail. (7)**